

UTKARSH SHARMA

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Research Interests: Physically-Based & Differentiable Rendering, Path Guiding, Neural Rendering, Inverse Graphics

Education

Indian Institute of Science (IISc)

Bangalore, India

M.Tech in Computer Science and Engineering | **CGPA: 9.1/10.0**

2024 – 2026

Key Coursework: Graphics & Visualization, 3D Vision & Inverse Graphics, Deep Learning for CV, Computational Topology, Optimization, ML for Signal Processing

Chandigarh University

Mohali, India

B.E. in Computer Science | **CGPA: 8.24/10.0**

2020 – 2024

Research Experience

M.Tech Thesis: Radiosity-Guided Accelerated Path Tracing | IISc

Jan – Apr 2026

- Advised by **Prof. Vijay Natarajan** and **Prof. R. Venkatesh Babu**; received the **Br. C. Selvam Endowment Award** for the best M.Tech project.
- Proposed a camera-independent path-guiding method using GPU progressive radiosity to find the top- K radiant contributors per point.
- Derived a closed-form, training-free von Mises–Fisher (vMF) mixture that turns these contributors into spherical sampling distributions.
- Fused CUDA kernel with bounded $\mathcal{O}(T)$ memory: up to $2.4\times$ higher sample throughput than NPM and PPG on diffuse scenes at comparable variance.

Selected Projects

Nabla: Differentiable Path Tracer | Python, PyTorch, CuPy/CUDA

May – Jul 2026

- Differentiable path tracer with a CUDA SAH BVH and autograd-aware intersections, giving gradients for geometry, materials, textures and camera.
- Implemented autodiff, Path Replay & Radiative Backpropagation (PRB/RB), warped-area and reparameterization estimators, all validated against finite differences.
- PRB/RB use up to $9.2\times$ less VRAM than autodiff (509 MB vs. 4,668 MB at 512^2); recovered textures and normal maps via inverse rendering.

Futaba: GPU Path Tracer | C++17, CUDA, OptiX, OpenGL

Mar – Aug 2025

- OptiX hardware path tracer (67–145 FPS on a 382K-triangle scene) with a software BVH fallback.
- Implemented NEE with MIS, volumetric media, Practical Path Guiding (SD-trees), env-map importance sampling, depth of field, and microfacet BSDFs.
- Interactive OpenGL viewport with G-buffer views and OptiX AI denoising.

Neural Path Guiding Framework | Mitsuba 3, Dr.Jit, PyTorch, Tiny-CUDA-NN

Jun – Dec 2025

- Implemented vMF mixtures, Neural Importance Sampling and Distribution Factorization in Mitsuba 3, reducing variance $2\text{--}4\times$.
- PyQt viewer for inspecting learned spherical PDFs; prototyped Neural Radiance Caching with hash grids.

Deep Learning Denoising for Monte Carlo Rendering | PyTorch, PBRT

Jan – Apr 2025

- Trained U-Net denoisers on low-spp PBRT renders with G-buffer inputs (albedo, normals, depth); evaluated with PSNR/SSIM.

Neural Radiance Fields & 3D Gaussian Splatting | PyTorch

Jan – May 2025

- Implemented NeRF and 3D Gaussian Splatting from scratch, reaching > 25 dB PSNR on NeRF Synthetic.

Technical Writing (usharma002.github.io)

Foundations of Physically Based Differentiable Rendering

Jul 2026

- From autodiff and Reynolds transport to edge sampling, reparameterization, warped-area and projective sampling, RB and PRB (6 papers, 2018–2023), with code in Nabla.

Importance Sampling & Path Guiding for Path Tracing

Dec 2025

- Monte Carlo and MIS foundations, then 10 guiding methods (2017–2025): PPG, Neural Importance Sampling, NRC, NPM, Distribution Factorization and more.

3D Reconstruction with 3DGS, NeRF, and Their Variants

Nov 2025

- 11 methods: radiance fields (NeRF, Plenoxels, TensorRF, Instant NGP, 3DGS) and sparse-view reconstruction (DUST3R, InstantSplat, FSGS).

Teaching Experience

Teaching Assistant, Computational Topology | IISc
Teaching Assistant, Algorithms & Programming | IISc

Jan – May 2026
Aug – Dec 2025

Skills

Languages: C++17, CUDA, Python, C, GLSL, SQL
Rendering & Graphics: Monte Carlo Path Tracing, Path Guiding, Differentiable Rendering, Volume Rendering, NeRF, 3DGS, NVIDIA OptiX, OpenGL
ML & Data: PyTorch, Tiny-CUDA-NN, U-Net, Neural Radiance Caching, Pandas, NumPy
Tools: Mitsuba 3, Dr.Jit, PBRT, CuPy, PyQt, CMake, Git
Spoken: Hindi (Native), English (Fluent)

Work Experience

Netradyne

Bangalore, India

Machine Learning Engineer, Data Science

July 2026 – Present

- Refine the **GreenZone[®] Score**, an AI-driven driver-performance metric, using large-scale telematics and video-derived data.
- Correlate driver-behavior signals with safety outcomes to improve predictive risk modeling.

Awards & Achievements

- **Br. C. Selvam Endowment Award** for the best M.Tech project, IISc (path guiding algorithms for Monte Carlo path tracing)
- **All India Rank 8** of 123,967 in GATE Computer Science 2024 (*top 0.01%*)
- **All India Rank 17** of 39,210 in GATE Data Science & AI 2024 (*top 0.05%*)
- **Winner**, CodeSmash 2.0 and **Finalist**, CodeRush 3.0 competitive programming contests